SIGNS AND TRIVIALITY CRITERIA FOR THE DEGREE OF AN INVERTIBLE SHEAF ON A PROPER CURVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- A section of an invertible sheaf on a curve is regular; characterization of O_X(D). Stacks Project, Divisors, Lemma divisors-lemma-characterize-OD (tag 0B3D). Realizes the invertible sheaf with a nonzero section as the sheaf associated to an effective Cartier divisor.
- Degree of an effective Cartier divisor and multiplicativity of the degree. Stacks Project, Varieties, Lemma lemma-degree-effective-Cartier-divisor (tag 0B3E) and Lemma lemma-degree-tensor-product (tag 0AYX). The first gives the sign and vanishing criterion and the second computes the degree of the twist used in the last part.

Source reference. Original reference 01X0: sources/stacks-chow/tags.txt, source lines 2501–2501. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AYY: sources/stacks-chow/tags.txt, source lines 13806–13806. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AYX: sources/stacks-chow/tags.txt, source lines 13805–13805. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a proper curve over k. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module.*

(1) *If $\mathcal{L}$ has a nonzero section, then* $\deg(\mathcal{L}) \geq 0$.
(2) *If $\mathcal{L}$ has a nonzero section s which vanishes at a point, then* $\deg(\mathcal{L}) > 0$.
(3) *If $\mathcal{L}$ and $\mathcal{L}^{-1}$ have nonzero sections, then $\mathcal{L} \cong \mathcal{O}_X$.*
(4) *If* $\deg(\mathcal{L}) \leq 0$ *and $\mathcal{L}$ has a nonzero section, then $\mathcal{L} \cong \mathcal{O}_X$.*
(5) *If $\mathcal{N} \to \mathcal{L}$ is a nonzero map of invertible $\mathcal{O}_X$-modules, then* $\deg(\mathcal{L}) \geq \deg(\mathcal{N})$ *and if equality holds then it is an isomorphism.*

Proof. Let s be a nonzero section of $\mathcal{L}$. Since X is a curve, we see that s is a regular section. Hence there is an effective Cartier divisor $D \subset X$ and an isomorphism $\mathcal{L} \to \mathcal{O}_X(D)$ mapping s the canonical section 1 of $\mathcal{O}_X(D)$, see Divisors, Lemma 01X0. Then $\deg(\mathcal{L}) = \deg(D)$ by Lemma 0AYY. As $\deg(D) \geq 0$ and $= 0$ if and

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only if $D = \emptyset$, this proves (1) and (2). In case (3) we see that $\deg(\mathcal{L}) = 0$ and $D = \emptyset$. Similarly for (4). To see (5) apply (1) and (4) to the invertible sheaf

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{N}^{\otimes -1} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{N}, \mathcal{L})$$

which has degree $\deg(\mathcal{L}) - \deg(\mathcal{N})$ by Lemma 0AYX. □